

Agri-Input Dealers and the Mango Farmer

How input costs get inflated, how products get mis-sold, and how wrong technical advice damages orchards, residues and incomes

Krishnagiri, Dharmapuri and Kaveripattinam belt, Tamil Nadu — with the legal remedies and the cost-control plan

Prepared: 18 August 2026 | **Third in series**, following the Agriculture & Horticulture Field Report and the Mango Value Chain Transparency Report
Intended users: Mango and vegetable growers and their FPOs; Assistant Agricultural Officers and Horticulture Officers; Insecticide and Fertiliser Inspectors; District Collectorate; consumer forums and civil-society monitors.

Read this before anything else.

1. Most input dealers are honest and many are genuinely useful. In a belt where public extension reaches only a fraction of farmers, the village dealer is often the only person who answers the phone at 9 p.m. when a hopper outbreak starts. This report is not an attack on the trade. It is a description of where the incentives point the wrong way, and what to do about it.

2. No firm, brand or individual is accused of anything here. Claims are marked as: **DOCUMENTED** (established in law, official data or press), **ALLEGED** (publicly reported but not judicially established), or **RISK** (a structurally possible practice that any control system must guard against, whether or not it is occurring locally).

3. The scale of the counterfeit problem is genuinely disputed, and this report says so. One industry body estimates almost 25% of India's pesticide market is counterfeit or illegal. Government sampling data placed in Parliament puts substandard samples at roughly 1.17%–2.7%, and RTI data from 28 states over eight years at about 2.45%. Both cannot be right. The likely explanation is that official sampling tests *labelled, branded stock on shelves* while the counterfeit trade runs through unlabelled, unbilled, off-book channels that inspectors rarely sample. **The honest position is that nobody knows the true figure — which is itself a finding.**

4. Nothing here is legal or agronomic advice for a specific orchard. Doses, molecules and intervals must come from a qualified source for your crop stage — see Part V.

Executive Summary

A mango grower in this belt spends money on inputs three to seven times a season, often more. Every one of those purchases is made from a person who **earns his living from the size of the sale** and who is, for most farmers, also the main source of technical advice. That single structural fact — **the seller is the advisor** — is the origin of nearly every problem in this report. It is not corruption. It is a conflict of interest that no amount of good intention fully removes.

The problem is made worse by an extension vacuum. National figures put one extension worker to roughly 1,162 operational holdings against a recommended 1:750 for irrigated and 1:1,000 for rainfed areas, with public extension reaching only a small minority of farmers. Where the state advisor is absent, the dealer fills the gap — and the advice that follows is, on average, advice to buy something.

Twelve mechanisms inflate the farmer's cost or damage his crop. They fall into three families:

- Billing and pricing** — sale without a cash memo (which destroys every later remedy at a stroke), charging above MRP, tie-in selling where a needed fertiliser is released only if a "tonic" is bought with it, credit at an interest rate never stated as an interest rate, and sale of expired or near-expiry stock.
- Product** — misbranded and substandard formulations, counterfeit packaging of reputed brands, cheaper adulterated product carried across state lines, and — the fastest-growing area — **unregistered "biopesticides", "tonics" and "growth promoters"**, an illicit trade one industry federation estimated at about ₹8,000 crore nationally in March 2026, with alleged adulteration by undeclared chemical pesticides and increasing sale through e-commerce.
- Advice** — calendar spraying instead of need-based spraying, cocktail mixes, the same active ingredient sold repeatedly under different brand names, fungicide sold for an insect problem, no pre-harvest interval guidance, unlabelled plant growth regulator dosing, and post-harvest ripening with a chemical banned under food safety law.

The residue consequence now has a price tag. Japan and Nepal have restricted Indian mango imports, and Nepal's suspension was reported as being over pesticide residues. In a belt already suffering a farm-gate price collapse to ₹4–5 per kg, a residue-driven loss of export outlets removes the one segment that pays well. **Every unnecessary spray a farmer is sold is therefore paid for twice — once at the counter and once at the ramp.**

The remedies already exist in law. The Insecticides Act, 1968 and Rules, 1971 license every seller and make it an offence to sell a misbranded insecticide. The Fertiliser (Control) Order, 1985 under the Essential Commodities Act, 1955 requires authorisation, cash memos and display of stock and price. The Legal Metrology Act governs weights. The Food Safety and Standards regulations prohibit calcium carbide. The Consumer Protection Act, 2019 allows a farmer to claim compensation for a defective product — and a district consumer forum has already ordered a manufacturer to compensate farmers and to withdraw a fungicide batch from the market. **All of these remedies collapse without one piece of paper: the cash memo.**

Part VI sets out how a mango grower in this belt can cut input spend by a realistic 20–40% without reducing protection — through soil testing, need-based spraying, subsidised bio-inputs available at KVK Dharmapuri at published rates, scheme assistance, and FPO bulk purchase.

Part I — Why the Dealer Became the Farmer's Agronomist

Condition	Consequence
Extension staff shortage	Roughly one extension worker per 1,162 operational holdings nationally, against recommended ratios of 1:750 (irrigated) and 1:1,000 (rainfed), with a large share of sanctioned posts unfilled and public extension reaching only a small minority of farmers. In Krishnagiri and Dharmapuri combined there are close to 4.9 lakh farmers.
Extension officers carry non-advisory duties	Input distribution, subsidy administration, quality control inspection, crop and insurance surveys, area reconciliation, data entry. Field advisory time is what gets squeezed.
Extension itself lacks market and sometimes practical currency	Published study of mango marketing in Krishnagiri found extension service providers themselves lacked information on changing market needs and could not advise producers appropriately.
The dealer is present, available and extends credit	Open six days a week, walking distance, gives product now and takes payment after harvest. To a marginal farmer with no cash in flowering season, this is not a shop — it is the bank and the doctor combined.
Low technical literacy among growers	A study of 250 mango growers found roughly 74% illiterate; 60% had no knowledge and 40% very little knowledge of the negative effects of excessive insecticide and fungicide use; growers were randomly using the same pesticide under different brand names , and frequently applying fungicides instead of insecticides — which of course killed no pests and did not prevent fruit drop.
Result	The person who decides what the farmer needs is the person who is paid by how much the farmer buys, on credit the farmer cannot easily refuse. This is the structural fault. Everything in Part II grows out of it.

Part II — Twelve Mechanisms of Cost Inflation and Mis-selling

A. Billing and pricing

Mechanism 1 — Sale without a cash memo **DOCUMENTED** as a regulatory concern

This is the master mechanism. Without a bill there is no proof of what was bought, from whom, at what price, of which batch. That single absence defeats: a misbranding prosecution, a Fertiliser Control Order case, a consumer complaint, a compensation claim, an insurance dispute, and any residue traceback. Every other malpractice in this report becomes easy once the bill is skipped — which is precisely why enforcement agencies advise growers on exactly this point. A Deputy Director of Law Enforcement in one state, after a drive that booked multiple dealers and recommended licence suspensions, gave orchardists and farmers a single piece of advice: **ask for a cash memo from the dealer when buying pesticides, fertilisers and seeds.**

Mechanism 2 — Charging above MRP or above the notified rate **RISK**

Printed MRP is enforceable. Risk points: charging above printed MRP in shortage periods; adding undisclosed "transport", "handling" or "loading" charges; selling a subsidised fertiliser at above the notified retail price; and short-weighing bags. Where no bill is issued, none of this is provable afterwards.

Mechanism 3 — Tie-in and linked selling **RISK**

The pattern: a scarce or price-controlled product — urea, DAP — is released only if the buyer also takes a high-margin, low-regulation item such as a micronutrient mixture, "growth tonic", soil conditioner or bio-stimulant. The controlled item carries the dealer almost no margin; the attached item carries most of it. The farmer records the purchase in his mind as "the cost of fertiliser", and never separates the two. **Under consumer law this is a restrictive or unfair trade practice, and it is actionable.**

Mechanism 4 — Credit whose interest is never named as interest **RISK**

Input taken in flowering season, settled after harvest six to eight months later. There is usually no stated interest rate; instead the credit price is simply higher than the cash price. A 15% difference over seven months is an effective annual rate in the mid-twenties; a 30% difference is far higher. Because it is never expressed as a percentage, the farmer cannot compare it against a Kisan Credit Card or a PACS loan — which is the point. Where credit also ties produce sale to the same person or an associate, the farmer loses on price as well as on cost.

Mechanism 5 — Expired, near-expiry and loose-repacked stock **RISK**

Expired formulations lose efficacy, so the farmer sprays, sees no result, and is sold a second product. Loose sale from a bulk container, decanted into an unlabelled bottle, removes the label — and with it the registration number, the batch, the expiry date, the dose, the pre-harvest interval and the antidote instructions. **A product without a label is not a pesticide; it is a liquid of unknown identity.**

B. Product quality and legality

Mechanism 6 — Misbranded and substandard formulations **DOCUMENTED**

Under the Insecticides Act, "misbranded" is defined in section 3(k) and covers, among other things, a product imitating or sold under the name of another insecticide, misleading or missing label content including registration number and warnings, toxicity higher than prescribed, or mixing or packing with a substance not covered by the registration. Government data placed in Parliament: over five years, 3,40,479 pesticide samples analysed, 3,639 prosecutions against firms involved in manufacture, sale or illegal import of substandard product, and 137 convictions. Documented enforcement examples include stock found with zero percent zinc against a specified 33 percent.

Mechanism 7 — Counterfeit product in genuine-looking packaging **DOCUMENTED** pattern

Fake fertilisers, seeds and pesticides sold in the packaging of branded companies have destroyed standing crops — in one documented instance, complaints from farmers in two districts were associated with damage across 3,000 acres of soybean, with affected farmers still servicing the loans taken to buy the inputs and receiving no compensation despite complaints to officials up to Collector level. **The cost of a counterfeit is never the price of the bottle. It is the season.**

Mechanism 8 — Cheap adulterated product moved across state borders **DOCUMENTED** pattern

A trade-side account from a border belt describes farmers crossing into a neighbouring state to buy an insecticide at ₹30-35/kg against a retail price of ₹100-105/kg — the price gap itself being the signal of adulteration. Enforcement teams in that state placed border districts on alert against trucks entering villages in the small hours selling fake product bearing the names of reputed manufacturers. Farmers bought them believing retail costs had simply been cut. **Relevance here is direct: this belt sits on the Karnataka and Andhra Pradesh borders, with heavy cross-border input and produce movement.**

Mechanism 9 — Unregistered "tonics", "bio-stimulants" and fake biopesticides **DOCUMENTED** — and the fastest-growing risk

In March 2026 an industry federation formally warned the Department of Agriculture and Farmers Welfare about the rapid proliferation of **unregistered biopesticides, alleged adulteration with chemical pesticides**, and the growing role of **e-commerce platforms** in facilitating their sale, estimating the size of this illicit trade at about **₹8,000 crore** nationally and describing the segment as expanding in a largely unchecked manner.

Why this category is the most dangerous one for a mango grower: "tonics", "growth promoters", "yield boosters" and "organic bio-stimulants" often sit outside the strict registration and labelling discipline that applies to a registered insecticide. That means: no CIB&RC registration number to verify, no declared active ingredient, no dose, no pre-harvest interval, no residue data — and if a chemical pesticide has been quietly added to make the product "work", **the farmer has applied an undeclared molecule to a fruit crop weeks before harvest and has no idea what will show up in a residue test.** These products also carry the highest dealer margins, which is exactly why they are pushed hardest.

C. Wrong advice

Mechanism 10 — Calendar spraying, cocktails, and the same molecule under many names **DOCUMENTED in mango growers**

Survey evidence from mango growers found insecticides and fungicides sprayed **2, 3, 4 and even more than 7 times** across flowering, green-fruit and mature stages; growers randomly using the same pesticide under different brand names; and fungicides applied where insecticides were needed, so pests were not killed and early fruit drop was not prevented. Each of these is a direct cost multiplier:

- **Calendar spraying** — spraying on a date rather than on a threshold means paying for sprays that were not needed.
- **Brand confusion** — buying the same active ingredient under three trade names means paying three times for one mode of action, while the pest population builds resistance to it.
- **Cocktail mixing** — three or four products in one tank, often with no compatibility basis, raising cost, phytotoxicity risk and residue load together.
- **Wrong target** — a fungicide against hoppers is money spent on an outcome that cannot occur.

Mechanism 11 — No pre-harvest interval advice, and the residue consequence **DOCUMENTED consequence**

The pre-harvest interval is the minimum time between the last application and harvest needed to keep residues below the Maximum Residue Limit. MRLs differ by destination country, so a spray that is fine for the domestic market can disqualify a consignment for export. A dealer selling a product close to harvest without stating the PHI has, in one transaction, sold a cost and created a liability.

This is no longer theoretical for this belt. Japan and Nepal have imposed restrictions on mango imports from India, with Nepal's suspension reported as being over pesticide residues. Krishnagiri mangoes have been exported to a large number of countries and export earnings have been significant. Losing residue-sensitive markets in the same seasons that farm-gate prices collapsed to ₹4–5 per kg removes the one outlet that pays a premium. **Residue discipline is now a price issue, not just a compliance issue.**

Mechanism 12 — Plant growth regulator and ripening chemical misuse **DOCUMENTED regulation + RISK in practice**

Paclobutrazol — a legitimate practice that is easy to sell wrongly. Paclobutrazol is used commercially as a soil application to counter mango's alternate-bearing habit, and it works: controlled trials on 22-year-old Alphonso recorded yield increases of 133% and 77% over control in on-years. The honest picture, from the same and related research, is more complex:

- It significantly retards plant height, spread and tree volume, reduces flush number and vigour, leaf number, leaf area and dry matter.
- Reduced dry matter and leaf area **combined with greater nutrient removal by the larger crop may weaken the tree significantly**, so treated trees need more careful nutrient management than untreated ones.
- It persists in soil — measured half-lives of roughly 58 to 77 days, with residues still detectable at 300 days, and other work describing persistence generally beyond a year.
- Applied continuously for more than one year it has been associated with reduced fruit size.
- On the reassuring side, in trials across a dose range up to 10 g a.i. per tree, **residues were not detected in fully mature fruits** harvested 85 days after fruit set.

The mis-selling risk is therefore not "paclobutrazol is bad". It is: dose escalation year on year because last year "worked"; sale without canopy-diameter-based dose calculation; no accompanying nutrition advice for a tree carrying a much heavier crop on less leaf area; and repeat annual application that mines the tree. A dealer paid per litre has no incentive to say "skip this year".

Calcium carbide ripening — outright illegal. Use of calcium carbide ("masala") is prohibited under Regulation 2.3.5 of the Food Safety and Standards (Prohibition and Restrictions on Sales) Regulations, 2011 for artificial ripening of mangoes, bananas, papayas and similar fruit. It releases acetylene gas carrying traces of arsenic and phosphorus, causing effects including dizziness, thirst, irritation, weakness, difficulty in swallowing, vomiting and skin ulcers, and it is hazardous to handlers. FSSAI ordered a nationwide enforcement drive in April 2026 with special drives in wholesale markets and immediate prosecution where banned substances are found. Ethylene gas is the permitted alternative at safe concentrations up to 100 ppm, and Ethephon 39% SL is endorsed by CIB&RC for uniform ripening — but **FSSAI has specifically flagged the practice of dipping fruits directly into ethephon solution, which violates the guidance prohibiting direct contact of fruit with ethylene in powder or liquid form.** Anyone selling a grower "masala" for ripening is selling a prosecution.

D. What it adds up to

Illustrative cost structure, one acre of bearing mango, one season. These figures are **illustrative arithmetic, not survey data** — use them as a template to fill in with your own bills, not as a benchmark.

Suppose a grower makes 6 spray rounds at ₹1,800 per acre per round in product = ₹10,800, plus a PGR application at ₹1,200, plus fertiliser and "tonics" at ₹6,000. Total input spend ≈ ₹18,000 per acre.

Now suppose that of the 6 rounds, **2 were calendar sprays not justified by any threshold** (–₹3,600), **1 round used a second brand of a molecule already applied** (–₹1,800), and **₹2,500 of the "tonic" spend was an unregistered product with no declared active ingredient** (–₹2,500). Nothing here involves counterfeit product or dishonesty by anyone — only advice that defaulted to selling.

Avoidable spend ≈ ₹7,900 per acre, or roughly 44% of input cost. On a crop being sold at ₹4–5 per kg, that is the difference between a thin margin and a loss. And the two removed spray rounds also lower residue load — which protects the export-grade price, if it returns.

Part III — Mango Crop Calendar: Where Advice Goes Wrong

Stage	Genuine risk at this stage	Common mis-selling / wrong-advice risk
Post-harvest pruning & rest (Jun–Aug)	Canopy management, sanitation, nutrition to rebuild reserves	Under-sold, because it involves labour and organic matter rather than chemicals. Yet this is the stage that determines next year's crop. A dealer has nothing to sell you here — which is exactly why the advice is missing.
Pre-flowering / PGR window	Alternate bearing; flowering induction decisions	Paclobutrazol sold without canopy-diameter dose calculation; dose escalated each year; repeat annual application; no accompanying nutrition plan for a tree carrying more fruit on less leaf area.
Flowering	Mango hopper, powdery mildew, anthracnose — these are real and can destroy a crop	Spraying to a calendar rather than to observed incidence; cocktail tank mixes; repeat use of one mode of action driving resistance; fungicide sold against an insect problem.
Fruit set & development	Fruit fly, stone weevil, nutrient demand peak	High-margin unregistered "fruit setters", "size boosters" and "shine tonics" with no declared active ingredient, no registration number and no residue data.
Pre-harvest (last 30–45 days)	Fruit fly, anthracnose management, harvest maturity judgement	The highest-stakes window. Any spray sold here without a stated pre-harvest interval creates residue risk. Ask for the PHI in writing on the bill, or do not buy.
Harvest & ripening	Maturity, careful handling, ripening	Calcium carbide ("masala") — prohibited; direct dipping of fruit in ethephon solution — contrary to FSSAI guidance. Ethylene gas in a proper chamber is the lawful route; zero-energy cool chambers have been evaluated as a low-cost option suited to small growers and traders.

The general test for any recommendation. Ask four questions before buying. **(1) What exactly am I treating — show me the pest or symptom on my tree. (2) What is the active ingredient and its registration number? (3) What is the dose per litre and the pre-harvest interval? (4) Will you write all of that on the bill?** A competent, honest dealer answers all four in thirty seconds. Hesitation on question four is the answer to the others.

Part IV — The Law the Farmer Can Use

Instrument	What it gives the farmer
Insecticides Act, 1968 and Insecticides Rules, 1971	Regulate import, manufacture, sale, transport, distribution and use of insecticides. No insecticide may be manufactured, stocked, sold or exhibited for sale without a licence. Section 3(k) defines "misbranded". Penalties apply for selling misbranded insecticides and for contravening licensing requirements. Insecticide Inspectors are empowered to inspect, seize stock (with a receipt in prescribed form) and draw samples; where a sample is taken, the Inspector must inform the person in writing and, in that person's presence, divide the sample into three parts — a safeguard that protects both farmer and dealer.
Fertiliser (Control) Order, 1985, under the Essential Commodities Act, 1955	Requires a Letter of Authorisation to deal in fertilisers, governs storage, and requires cash memos and display of stock and price. Documented enforcement uses FCO clauses together with Essential Commodities Act sections 3(2) (d) and 7 — an FIR in 2026 booked a dealer and a supplying firm under exactly this combination alongside Insecticides Act sections 13, 17, 18 and 29 and provisions of the Bharatiya Nyaya Sanhita, for sourcing and selling product to farmers without valid licence or authorisation and storing chemicals in a residential area.
Consumer Protection Act, 2019	A farmer is a consumer. The three-tier District, State and National Commissions deal with defective goods, deficient services, unfair trade practices, restrictive trade practices, overcharging by traders and hazardous goods . Precedent exists: a District Consumer Disputes Redressal Forum ordered that a named fungicide batch be immediately removed from the market and that the company compensate affected farmers — with compensation actually paid to a first tranche of complainants. This is the most under-used remedy available to Indian farmers.
Food Safety and Standards Act, 2006 and the Prohibition and Restrictions on Sales Regulations, 2011	Regulation 2.3.5 prohibits calcium carbide as an artificial ripening agent. FSSAI has directed state Food Safety Commissioners to run special drives in wholesale markets with immediate prosecution where banned substances are found, and has flagged improper direct dipping of fruit in ethephon solution. Consumers and growers are urged to report misuse to the State Commissioner of Food Safety.
Legal Metrology Act, 2009	Governs weighing and measuring instruments and net-quantity declarations — applies to short-weight fertiliser bags as much as to a mandi weighbridge.
Bharatiya Nyaya Sanhita	Cheating and criminal breach of trust where dishonest intent in the sale of a fake product can be shown.

An important nuance that cuts both ways — read it before naming a dealer in a complaint.

A High Court has held that a marketing agent or dealer who merely **traded** in a product, where the sample was drawn from a **sealed packet** and there was no allegation of improper storage, **cannot be held vicariously liable for misbranding** — because section 3(k) misbranding relates to the label and its contents, and the statute intends to penalise the person who committed the offence, not those merely dealing in a product over whose quality and content they had no control.

What this means practically: if a dealer sold you a sealed, properly labelled pack of a branded product that later tests substandard, your claim probably runs against the **manufacturer**, not the shopkeeper. The dealer's own liability arises where **he** did something — sold without a licence, sold loose or unlabelled, sold expired stock, stored improperly, sold an unregistered product, refused a bill, or overcharged. Directing a complaint at the wrong party is the commonest reason these cases fail.

Reform in progress. The Pesticide Management Bill, 2020 — intended to replace the 1968 Act — remains pending. The Standing Committee on Agriculture identified the 1968 Act's weaknesses as a restrictive definition of "insecticide", **absence of residue tolerance limits as a pre-condition for registration**, and inadequate penalties. Separately, the central government has been reported as drafting a comprehensive law against counterfeit seeds, pesticides and fertilisers with stronger penalties including criminal prosecution and licence revocation. Note also that the term "spurious" has **no formal definition** under the Insecticides Act — only "misbranded" — which is a real obstacle to prosecution and worth knowing before filing.

Part V — The Farmer's Purchase Protocol

5.1 Before you buy — separate the diagnosis from the sale

This is the single highest-value habit in this entire report. **Get the diagnosis from someone who does not sell the cure.** In this belt that means:

- The **Assistant Agricultural Officer** on his scheduled village visit — and if you do not know his visit day, ask at the Agricultural Extension Centre for the published calendar.
- **TNAU Regional Research Station, Paiyur**, in Kaveripattinam union — functioning since 1973 and the correct technical authority for this agro-climatic zone. It is the most under-used asset in the block.
- **KVK Dharmapuri** — on-farm testing, front-line demonstrations and training.
- The **Horticulture Officer** for scheme-linked and orchard-specific questions.

Then go to the dealer with a decision already made, and buy that. A dealer asked "what should I spray?" will answer with a product. A dealer asked "do you stock this molecule at this concentration?" will answer with a price.

5.2 At the counter — the eight checks

1. **Licence displayed?** Selling without a licence is itself an offence.
2. **Sealed, labelled original pack?** Never accept loose product decanted into an unlabelled bottle.
3. **Registration number on the label?** No CIB&RC registration number means it is not a registered pesticide, whatever the shop says it does.
4. **Manufacturing and expiry date** — check both, every time.
5. **Batch number** — this is what makes a later complaint traceable.
6. **MRP printed** — pay that, not more.
7. **Dose and pre-harvest interval stated** — ask for both to be written on the bill.
8. **Printed cash memo, with the shop's name, licence number, product name, batch, quantity, rate and total.** No bill, no purchase. This is not negotiable.

5.3 After you buy — keep the evidence pack until harvest is sold

- Keep **the bill, the empty container and the label** together in one place — not the container alone, not the bill alone.
- Photograph the label and batch number on your phone the day you buy.
- Maintain a **spray register**: date, block, crop stage, problem observed, product, dose, quantity, and the harvest date it permits. Annexure A gives the format.
- Keep bank payment records where possible rather than cash.
- If a product fails or damages the crop: **do not destroy anything**. Report immediately to the Assistant Director of Agriculture / block office so an inspector can draw a sample before the stock moves, photograph the damage with dates, and get a field inspection recorded in writing.

5.4 Escalation ladder

Step	Where	For what
1	Assistant Director of Agriculture, block Agricultural Extension Centre	First report of suspected fake or substandard input; request sampling by the Insecticide Inspector; field damage inspection
2	Insecticide Inspector / Fertiliser Inspector (block-level, usually the block agriculture officer)	Sampling, seizure, prosecution for misbranding, unlicensed sale, expired stock
3	Joint Director of Agriculture (district)	Licence suspension or cancellation; inaction at block level
4	Pesticide Testing Laboratory (regional)	Analysis of pesticide samples for composition; misbranded results go back to block Insecticide Inspectors for legal action
5	District Consumer Disputes Redressal Commission	Compensation for defective goods, unfair or restrictive trade practice, overcharging. Requires the bill.
6	State Commissioner of Food Safety / district Food Safety Officer	Calcium carbide ripening; improper ethephon dipping
7	Legal Metrology Department	Short weight, incorrect net-quantity declaration
8	District Collector — monthly Agricultural Grievance Day	Administrative failure; no action on earlier complaints
9	Directorate of Vigilance and Anti-Corruption	Where a public servant is soliciting or accepting gratification, or shielding a dealer
10	Police / court	Cheating; organised counterfeit supply

Part VI — Cutting Input Cost Without Cutting Protection

A realistic 20–40% reduction in input spend is achievable in this belt through the following, none of which requires accepting more crop loss.

Measure	How it saves money
Soil test first, fertilise second	The Soil Health Card mission uses GPS sampling on a 2.5 ha grid in irrigated areas and 10 ha grid in rainfed areas, giving crop-specific fertiliser recommendations. Most over-application in orchards is of nutrients already sufficient in the soil. A test costs almost nothing and is in any case required documentation for horticulture scheme components — so front-load it.
Need-based spraying, not calendar spraying	Scout, then spray. Agro Eco System Analysis based crop health management is a funded scheme component. Two avoided rounds per season is typically the largest single saving available.
One molecule, one purpose, rotated	Know the active ingredient, not the brand name. This ends double-buying the same chemistry under two labels and slows resistance build-up — which is itself a future cost.
Subsidised bio-inputs from KVK	KVK Dharmapuri has published rates for inputs including Trichoderma viride at ₹168/kg, vermicompost at ₹12/kg and azolla seed at ₹60/kg , alongside crop boosters and salt lick. Buying bio-agents from the institution rather than from a shop removes both the margin and the counterfeit risk.
Take the IPM/INM scheme assistance you are entitled to	Published district scheme rates include INM micronutrient distribution at ₹500/ha, bio-fertiliser at ₹300/ha, IPM plant protection chemicals and bio-agents at ₹500/ha, bio-control agents at ₹500/ha, and battery-operated or power sprayers at ₹3,000–3,800. Horticulture schemes fund organic cultivation incentives and beekeeping (colonies at 40% subsidy, ₹1,600 per colony) — and bees in a mango orchard improve fruit set while producing a second income.
Buy through the FPO, not individually	The FIG-FPG-FPO structure — 20 farmers per FIG, five FIGs to a 100-farmer FPG — is explicitly designed for collective purchasing of inputs alongside collective cultivation and market linkage. A 100-farmer order buys at a different price, from an authorised distributor, with a single verifiable invoice, and removes the credit dependency that drives most mis-selling.
Break the input-credit trap	Compare dealer credit against a Kisan Credit Card or PACS loan by converting the cash-versus-credit price gap into an annual percentage. Once expressed as a rate, most input credit is visibly the most expensive money on the farm.
Refuse unregistered "tonics"	No registration number, no declared active ingredient, no dose, no PHI — no purchase. This category carries the highest margin and the weakest evidence base, and is where the ₹8,000 crore illicit trade sits.
Manage residues deliberately	Maintain a spray register with pre-harvest intervals and set a hard cut-off date before harvest. This protects access to residue-sensitive markets — which, given the restrictions imposed by Japan and Nepal, is now a price question.

Part VII — What the System Should Do

#	Measure	Detail
1	Printed cash memo made an enforced licence condition	Bill must carry shop name and licence number, product, batch, mfg/expiry, quantity, MRP, dose and PHI. Suspend the licence for repeated non-issuance. This one measure enables every other remedy in Part IV.
2	Stock and price display board at every input shop	Daily updated, in Tamil, showing available stock and rates — already contemplated under fertiliser control provisions and rarely enforced.
3	Publish the licence register	So a farmer can verify before buying whether the shop is licensed for pesticides, for fertilisers, or for neither.
4	Publish sampling and test results, and enforcement outcomes	Samples drawn, results, prosecutions launched, licences suspended — published quarterly at district level. A published suspension changes behaviour district-wide.
5	Pre-season sampling drives with officers stationed at key sales points	Documented practice in other states: officers stationed at key sales points during peak season, rigorous sampling at retail and distribution level, licences suspended or revoked where fake product is found.
6	Special enforcement focus on the unregistered "bio-stimulant" and e-commerce channel	This is where the growth is and where registration discipline is weakest. Sampling regimes designed around branded shelf stock will keep missing it.
7	Border-district vigilance	This belt adjoins two states with documented cross-border movement of cheap adulterated product. Joint checks with Revenue and Police on night movement into villages.
8	Certify and train dealers as advisors — then hold them to it	Structured dealer certification in agri-extension, plus periodic anti-counterfeit training. If the dealer is going to be the advisor in practice, make him a qualified and accountable one. Industry-government programmes have already written to lakhs of dealers on counterfeit risk and sanctions.
9	Rebuild independent advisory presence	Fill AAO vacancies, publish the FOIAES fortnightly visit calendar so farmers know when free advice arrives, and equip every field officer with current pest surveillance and residue guidance. The dealer's advisory monopoly is a symptom of the extension vacuum; close the vacuum and the symptom shrinks.
10	Accessible residue testing	Affordable, quick residue testing available to growers and FPOs before harvest, so a residue problem is found in the orchard rather than at a port.
11	Village-level input awareness during the visit circuit	The AAO's existing group meetings are the natural place to teach label reading, registration numbers, PHI and bill-keeping — at zero marginal cost.

12	Publicise the consumer forum route	Farmers do not use it because they do not know it applies to them. A single well-publicised district case, as happened elsewhere, changes purchasing behaviour across a season.
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Annexure A — Input Purchase and Spray Register

Field	Entry
Date of purchase	
Shop name, address, licence number	
Bill number (printed cash memo obtained? Y/N)	
Product trade name	
Active ingredient and concentration	
CIB&RC registration number (or "none stated")	
Batch number / manufacturing date / expiry date	
Quantity purchased / MRP / rate charged / total paid	
Cash or credit; if credit, cash price vs credit price and settlement date	
Who diagnosed the problem (AAO / KVK / RRS Paiyur / dealer / self)	
Problem actually observed on the crop	
Date of application / block or acreage sprayed / crop stage	
Dose used per litre and total volume applied	
Pre-harvest interval stated / earliest permitted harvest date	
Result observed after 7–10 days	
Container and label retained? (Y/N)	

Annexure B — Red-Flag Checklist at the Counter

Any two of these together should stop the purchase.

- No printed cash memo offered, or "I'll write it in the notebook".
- Product decanted loose into an unlabelled bottle or bag.
- No registration number, or no active ingredient declared on the label.
- No manufacturing date, expiry date or batch number.
- Price far below the normal market rate for that molecule.
- Packaging that looks like a reputed brand but with spelling, printing or seal irregularities.
- A "tonic", "booster" or "setter" recommended with no explanation of what it contains.
- A fertiliser released only if another product is bought with it.
- Dose or pre-harvest interval not stated, or refused when asked in writing.
- A recommendation given without the dealer ever seeing the affected tree, leaf or fruit.
- Three or four products recommended to be mixed in one tank with no compatibility basis.
- "Masala" or any carbide product offered for ripening — this is prohibited, and buying it exposes you, not only the seller.

Annexure C — Complaint Format for a Suspected Fake or Substandard Input

Field	Entry
Complainant name, village, mobile	
Crop, variety, area affected, tree age	
Shop name, address, licence number	
Bill number and date (attach original or copy)	
Product name, active ingredient, registration number, batch, mfg/expiry	
Quantity purchased and amount paid	
Date, dose and method of application; crop stage	
Damage or failure observed, with dates	
Estimated loss (area × expected yield × prevailing rate)	
Evidence attached: bill, label, empty container, dated photographs, spray register extract, witness names	

Was a sample drawn by the Insecticide/Fertiliser Inspector? Date and reference	
Field inspection conducted by which officer, on which date	
Complaint directed against: manufacturer / dealer / both (see the vicarious-liability note in Part IV)	
Relief sought	
Submitted to / date / acknowledgement number	

Sources and Caveats

Legal sources: Insecticides Act, 1968 and Insecticides Rules, 1971 (including section 3(k) misbranding and Inspector powers of seizure and three-part sampling); Fertiliser (Control) Order, 1985 and Essential Commodities Act, 1955; Consumer Protection Act, 2019 and the three-tier redressal machinery; Food Safety and Standards Act, 2006 with the Prohibition and Restrictions on Sales Regulations, 2011 (Regulation 2.3.5) and FSSAI guidance on ethylene as a fruit ripener; Legal Metrology Act, 2009. Pending reform: Pesticide Management Bill, 2020 and Standing Committee on Agriculture observations, via PRS Legislative Research. High Court authority on the limits of dealer vicarious liability for misbranding of sealed branded product.

Data and reporting: Government replies in Parliament on pesticide sample analysis, prosecutions and convictions; industry federation estimates on counterfeit market share and on unregistered biopesticides (March 2026); RTI-based sampling data across 28 states; documented enforcement drives and FIRs in other states; FSSAI directives of 2024 and April 2026 on calcium carbide and ethephon misuse; peer-reviewed work on paclobutrazol effects on mango yield, tree vigour, soil persistence and fruit residues; survey research on mango growers' pesticide knowledge and practice; reporting on Japan and Nepal restrictions on Indian mango imports; KVK Dharmapuri published input rates; Krishnagiri and Dharmapuri District Administration scheme pages.

Caveats. Subsidy rates and input prices are as published in recent district and KVK material and are revised periodically — verify current rates before relying on them. The cost example in Part II is **illustrative arithmetic, not survey data**. The magnitude of the counterfeit input problem is genuinely contested and this report presents both the high industry estimate and the low official sampling figures rather than choosing between them. Agronomic points on paclobutrazol, ethephon and pesticide choice are drawn from published research and regulatory guidance and are included to inform the mis-selling discussion — **they are not a dose recommendation for any orchard**. For crop-specific recommendations consult TNAU Regional Research Station Paiyur, KVK Dharmapuri, or the Horticulture and Agriculture departments. For any formal complaint or claim, take advice from a qualified lawyer. Nothing in this report alleges wrongdoing by any identified person, dealer, manufacturer or brand.